Problem Set 1

Starter

1 (E)**.** Let A and B be nonempty subsets of real numbers. Show that:

(a) $\sup(-A) = -\inf A$;

(b) $\sup(A+B) = \sup A + \sup B$;

(c) $\sup(A \cup B) = \max\{\sup A, \sup B\}$.

2 (E)**.** (a) Show that given any number $x \in \mathbb{R}$, there exists an $n \in \mathbb{N}$ such that $n > x$.

(b) Show that given any real number $y > 0$, there exists an $n \in \mathbb{N}$ such that $\frac{1}{n} < y$.

3 (L-M)**.** Let $u \in \mathbb{R}$ be an upper bound for a nonempty subset $A \subseteq \mathbb{R}$. Show that $u = \sup A$ if and only if for all $\varepsilon > 0$, there exists an element $a \in A$ satisfying $u - \varepsilon < a$.

4 (L-M)**.** Determine the least upper bound and the greatest lower bound of the set

$$A = \{\sqrt{n} - \lfloor \sqrt{n} \rfloor : n \in \mathbb{N}\}.$$

Hint: Note that $(k-1)^2 < k^2 - 1 < k^2$ for all integer $k \geq 2$.

Advanced

5 (M)**.** Let $A = \{x \in \mathbb{Q} : x > 0,\ x^2 < 2\}$. Show that A does not have supremum in $\mathbb{Q}$.

6 (M)**.** Prove that if $a > 0$, $k \in \mathbb{N}$,

$$H_- = \{x > 0 : x^k < a\},$$
$$H_+ = \{x > 0 : x^k > a\},$$

then $(\sup H_-)^k = (\inf H_+)^k = a$. (It shows the existence of kth root of a.)

7 (M)**.** Show that $\mathbb{Q}$ is dense in $\mathbb{R}$, i.e., for all $a, b \in \mathbb{R}$ with $a < b$, there exists $r \in \mathbb{Q}$ satisfying $a < r < b$. Deduce that $\mathbb{Q}^{\complement}$ is dense in $\mathbb{R}$.

8 (H)**.** Show that for any irrational number α and for any positive integer n there exist a positive integer q_n and an integer p_n such that

$$\left|\alpha - \frac{p_n}{q_n}\right| < \frac{1}{nq_n}.$$

Show also that $\{p_n\}$ and $\{q_n\}$ can be chosen in such a way that

$$\left|\alpha - \frac{p_n}{q_n}\right| < \frac{1}{q_n^2}.$$